

The cyborg is a convergent form, and he has the tools to defy death

The Cyborg

We've spoken about cyborgs earlier, in the January 2005 issue of *Digit*. We laid out a definition, and spoke about what kind of research is going on in the field of cybernetics, which, according to Dictionary.com, is "the theoretical study of communication and control processes in biological, mechanical, and electronic systems, especially the comparison of these processes in biological and artificial systems."

'Cyborg' happens to be short for 'cybernetic organism', and a cyborg, with his (or her) implants and wirings-up, is neither purely human nor purely machine; he is a convergent form, and he has the tools to defy death.

Apart from just living forever as cyborgs, we could have greatly enriched lives. No more will our minds, which can traverse vast lengths of space and time in the blink of an eye, be bogged down by the physical limitations of speech, language, touch, sight and other boundaries.

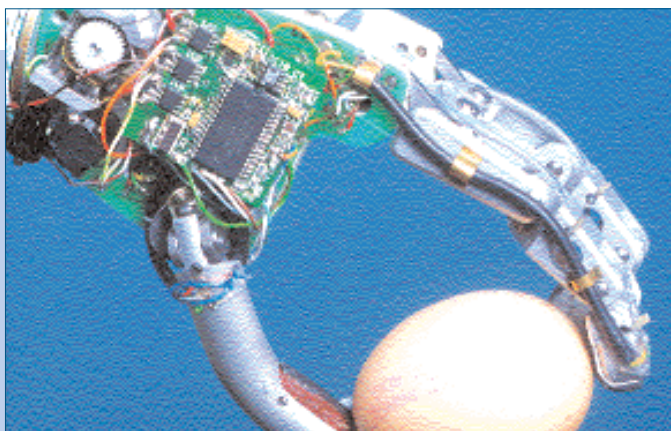
Wired Humans

Strictly speaking, anyone with an implant of some sort may be called a cyborg, because his or her body has been augmented. That includes people with pacemakers and artificial limbs. An extreme example of this is clothing—it can be considered an augmentation of the skin.

However, we must remember that what defines a cyborg is that his relations with the universe have changed, usually for the better. Seen in that light, while it is true that someone with an artificial limb has a slightly different interface with the universe, he is not, in a subjective sense, a 'true cyborg'. He experiences, and responds to, reality pretty much the same way we non-cyborgs do. What, then, do we mean by a 'true cyborg'?

We may define a true cyborg as one whose relations with the universe have changed in a dramatic way; as one whose body has merged, quite literally and very significantly, with the mechanical.

So what kind of creature are we imagining? We can borrow from science fiction and talk about limbs that can be torn apart and replaced, eyes that can pop out of their sockets and have the capability of zooming in or out, and infrared or ultraviolet vision. But we can go further, and imagine a being with add-ons that give him tremendous powers.



There is nothing, conceptually speaking, to prevent us reaching a state where we augment all our body parts—limbs and internal organs included

For example, this creature would have transmitting and receiving antennas wired to its brain through which it could communicate directly via thought; most of its body parts and internal organs would be replaceable, mechanical and synthetic; its senses would be augmented in such a manner that it takes in reality in richer colours than we do; its implants would give him memory and brain power far superior to ours. It is this organism we are calling the true cyborg.

Cyborg Experiences

So how would you spend your time as a cyborg? The same way you do now, except that your reality is enhanced by a variety of implants and accessories.

All unpleasantness is done away with, and only desirable experiences pass on through to your brain. You could have a special interface, exclusively through which you communicate with the world. This interface might filter out unwanted people and conversations; the interface could also enable the recording of your thoughts and experiences so that you can play them back later. You could actually re-live your favourite experiences.

Think about what defines our lives today...

Books and movies? As a cyborg, you could 'read' and 'view' via your thought receiver antenna, from a central server common to all of cyborgdom. This server would record the experiences of all fellow cyborgs, so there is a much vaster pool of information to draw from than what we have now. And the transmission would be instantaneous, doing away with the need for spending hours reading a book or watching a movie.

Social interaction? Already, many people prefer interaction over the Internet, rather than face to face. In cyborgdom, there would be no need for any unwanted interactions; entities could be screened for suitability, and interactions would be at light-speed, without the need for elaborate, time-consuming social customs.

Sex, and other pleasures? If, or when, you felt the desire for such experiences, you could simply tap into your experience server, and have the experience delivered—either as a standalone experience 'applet', or in conjunction with a fellow cyborg. As a matter of fact, there need be no males or females in cyborgdom; all entities would be equal.

Work? Ah, that lowest common denominator of all human endeavour! Well, there would be no work required of the cyborg version of you, since work only serves to support the being—and in the stage of evolution we're talking about, life-support systems, Matrix-style, would be the norm.

In essence, your life as a cyborg would revolve around experience; you create your own experience-space, think and live in it, and are isolated from nature. You are no longer bound by space and time.

Ethics

According to Kevin Warwick, professor of cybernetics at the University of Reading, UK (see his column *Human-machine Convergence* on page 86), it is possible to control nearly every aspect of our psychology by the means of implants. Stage fear? Rely on the implant to get rid of it.

In such a scenario, there is bound to be a rush for these mind and body-altering augmentations. Will human qualities mean anything any more?